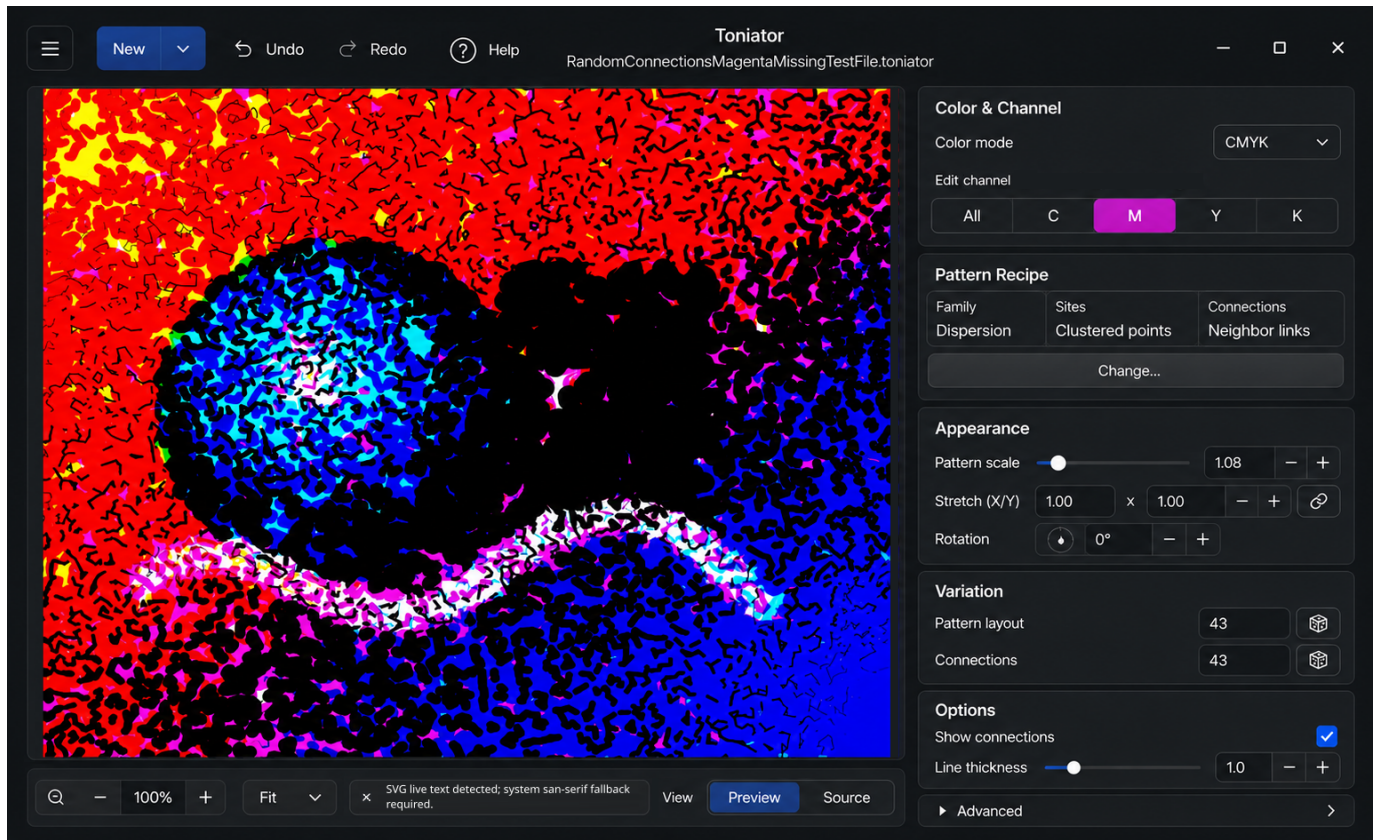


Toniator Main UI Specification v1.2

Final main-window mockup revision. Pure GTK4 target. Reference baseline: MainWindowMockup(1).png.



1. Purpose and design intent

Purpose. Define the main Toniator editing window shown in the final mockup. The canvas is the dominant workspace; the right inspector contains the editing controls; secondary file/application actions are consolidated rather than exposed as persistent navigation.

Primary workflow. Open or create a document, choose the edit channel, inspect or change the pattern recipe, adjust appearance/variation/options, compare Preview with Source, then save or export using the appropriate command.

Progressive disclosure. Frequently used artist-facing controls remain visible. Specialist controls stay under the collapsed Advanced section.

2. Main window regions

Region	Required behavior
Header bar	Hamburger menu; New split/dropdown action; Undo; Redo; Help; centered application/document identity; native window controls.
Canvas	Largest region. Displays either the generated Preview or the Source image.
Bottom canvas bar	Zoom controls, Fit control, inline status/warning area, and Preview / Source view toggle.
Right inspector	Top-down editing surface: Color & Channel, Pattern Recipe, Appearance, Variation, Options, Advanced.

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Header, menu, canvas, and channel-editing behavior.

3. Header bar and hamburger menu

Element	Specification
Hamburger menu	Home for secondary file/application actions. Confirmed examples include Open, Recent, Save, Save As..., Export, and About. This list is illustrative rather than an exhaustive menu contract.
New	Persistent split/dropdown action as shown in the mockup. Exact dropdown contents are outside this main-window specification.
Undo / Redo	Persistent history controls. Disabled when the corresponding action is unavailable.
Help	Persistent direct-access Help control in the header.
Application / document identity	Centered two-line identity: application name (Toniator) above the current document filename.
Window controls	Native minimize, maximize/restore, and close controls at the right edge.

Duplication rule. The main window does not add a persistent left sidebar or duplicate menu-only actions as navigation modes.

4. Canvas and bottom canvas bar

Control	Behavior
Canvas	The artwork occupies the available central area and remains the visual priority of the window.
Zoom out / percentage / zoom in	Standard zoom controls. The percentage reflects current zoom and may be directly adjustable if supported by the existing implementation.
Fit	Fits the source/preview to the available canvas viewport. The dropdown affordance may expose existing fit variants; this specification does not add new variants.
Status / warning area	Inline, non-modal message area between Fit and View. The final mockup shows the live-SVG-text warning here with a dismiss control. Messages should not become a separate permanent panel.
View: Preview / Source	Exclusive two-state selector. Preview shows Toniator output; Source shows the input image for visual comparison. Preserve viewport position/scale when practical so switching views is spatially meaningful.

5. Color & Channel

This is the first inspector section because it establishes both the document color model and the target of subsequent channel-scoped pattern edits.

Control	Behavior
Color mode	Dropdown for the document color model. Changing the model must immediately rebuild the Edit channel selector to match the channels provided by that model.
Edit channel - CMYK	Exclusive segmented buttons: All, C, M, Y, K.
Edit channel - RGB	Exclusive segmented buttons: All, R, G, B.
Other color models	Use the actual editable channels defined by the selected color model. Do not hard-code CMYK/RGB channel assumptions into the selector.
Channel highlight	The selected channel button uses a highlight that visually corresponds to that channel color (for example C/cyan, M/magenta, Y/yellow, K/black; R/red, G/green, B/blue). Text/icon contrast must remain readable. All uses a neutral/system selection treatment.
Edit scope	Selecting one channel makes subsequent channel-scoped pattern changes target that channel. All targets all applicable channels. Channel selection is an edit target, not a preview-isolation control.

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Pattern recipe and artist-facing editing controls.

6. Pattern Recipe

The Pattern Recipe section replaces redundant "Pattern" / "Pattern type" wording with a concise description of how the current pattern is constructed.

Field	Meaning in the main inspector	Mockup example
Family	Broad pattern family.	Dispersion
Sites	Site/point strategy.	Clustered points
Connections	Connection strategy.	Neighbor links
Change...	Single main-window entry point for changing the current recipe. Opens the existing pattern-selection workflow.	Button below summary

Channel relationship. When a single Edit channel is selected, the recipe summary and recipe edits correspond to that edit target. When All is selected and channels do not share identical recipe state, the UI must not falsely imply that they do; the exact mixed-value presentation follows the application's existing authority model.

7. Appearance

Control	Main-window presentation
Pattern scale	Slider plus numeric value and compact decrement/increment controls as shown.
Stretch (X/Y)	Two numeric values, one for X and one for Y, with a link control for coupled editing.
Rotation	Angle control with compact dial/indicator, numeric degree value, and decrement/increment controls.

8. Variation

Control	Behavior
Pattern layout	Deterministic layout variant value with a randomize action (dice icon in the mockup). Primary UI does not need to expose the implementation term "seed".
Connections	Independent deterministic connection variant value with randomize action when the current recipe supports that concept.

9. Options and Advanced

Section	Behavior
Options	Compact, immediately understandable controls that apply to the current recipe. The final mockup shows Show connections and Line thickness. Controls should appear only when applicable to the active recipe/edit target.
Advanced	Collapsed by default. Contains the more technical controls already supported by the application/descriptor model. It is progressive disclosure, not a second authority for values already shown above.

10. Explicit non-goals of this main-window spec

- No persistent left sidebar or Design/Channels navigation modes.
- No duplicate Pattern Library button in the main window.
- No "Preview: All channels" selector under the canvas.
- No assumption that selecting an Edit channel isolates the displayed preview.
- No exhaustive definition of the hamburger-menu contents, Pattern Change workflow, Export UI, or Advanced controls beyond what is required to place them correctly in the main-window hierarchy.

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Interaction, responsive behavior, terminology, and GTK4 constraints.

11. Interaction rules

- Edit channel is authoritative for channel-scoped edits. Changing C/M/Y/K or R/G/B changes where subsequent applicable pattern changes are written; it does not change the View selector.
- Preview and Source are view states only. They do not change the edit target, recipe, or document settings.
- Color-mode changes are structural UI changes. Rebuild the Edit channel button set immediately and update channel-colored selected states to match the new model.
- Inspector sections are contextual. Controls that do not apply to the current recipe/channel should not occupy permanent space merely to expose capability.
- No hidden duplicate navigation. Pattern changing occurs through Change... in Pattern Recipe; file/application actions belong in the header or hamburger menu as shown.

12. Responsive behavior

Area	Requirement
Right inspector	Scrollable vertically when its content exceeds window height. Preserve section order.
Horizontal allocation	Canvas receives the flexible share of width. Inspector maintains a usable minimum width for labels, segmented channel buttons, and numeric controls.
Narrow windows	Do not introduce a permanent left navigation rail. If the inspector must be temporarily hidden/collapsed at very narrow widths, that behavior should preserve the same inspector hierarchy rather than creating a second navigation model.
Color & Channel	Segmented Edit channel selector adapts to the selected color model and available width while keeping all channel labels explicit.

13. Terminology

Use	Meaning / reason
Color mode	Document color model.
Edit channel	Target for channel-scoped pattern editing.
Pattern Recipe	Family / Sites / Connections summary.
Appearance	Artist-facing size, stretch, and rotation controls.
Variation	Reproducible layout/connection variants without foregrounding seed jargon.
Pattern scale	Overall pattern size. Use density only for a control that literally changes marks/sites per unit area.

14. GTK4 implementation constraints

The mockup is intentionally implementable with standard GTK4. Widget choices below are implementation examples, not a requirement to replace equivalent controls already present in Toniator.

Constraint	GTK4 guidance
Pure GTK4	No libadwaita dependency is required by this specification.
Header / menu	GtkHeaderBar or equivalent GTK4 composition; GtkMenuButton plus GMenuModel/GAction is appropriate for the hamburger menu.
Main layout / inspector	GtkPaned or GtkGrid for canvas + inspector; GtkScrolledWindow for vertically stacked inspector sections.
Controls / disclosure	Use standard GTK4 dropdown, grouped toggle, scale, spin, check, button, and expander/revealer behaviors as appropriate.
Accessibility	Channel color is never the only identifier. Preserve text labels/accessibility names, tooltips for icon-only controls, and logical keyboard focus order.